Experiment 2: Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

1. Aim

To model and analyze a 50 Ω microwave transmission line using MATLAB, determine the scattering parameters \(S_{11}\), \(S_{21}\), \(S_{12}\), and \(S_{22}\), and evaluate return loss, insertion loss, transmission efficiency, and optimum operating frequency over the 8–12 GHz frequency range.

2. Objectives

1. To develop a mathematical model of a microwave transmission line operating at microwave frequencies.
2. To determine the characteristic impedance, propagation constant, and transmission characteristics of the selected structure.
3. To obtain and analyze the S-parameters \(S_{11}\), \(S_{21}\), \(S_{12}\), and \(S_{22}\) over a specified frequency range.
4. To evaluate return loss, insertion loss, and transmission efficiency from the S-parameter response.
5. To identify the optimum operating frequency based on minimum reflection and maximum transmission.

3. Software Required

MATLAB

4.Program Coding:

```
clc; clear; close all;


% Input parameters
f = 8:0.05:12;            % Frequency in GHz
f0 = 10;                  % Center frequency
Z0 = 50;                  % Reference impedance (ohms)
L = 50e-3;                % Length in metres


% Illustrative two-port model
S11dB = -25 - 15*(f-10).^2;
S21dB = -2 - 0.1*(f-10).^2;
```

```
% Linear S-parameter magnitudes
S11 = 10.^(S11dB/20);
S21 = 10.^(S21dB/20);

% Performance calculations
RL = -20*log10(abs(S11));   % Return loss (dB)
IL = -20*log10(abs(S21));   % Insertion loss (dB)
Eff = abs(S21).^2*100;      % Transmission efficiency (%)

% Find values at 10 GHz
[~,i0] = min(abs(f-f0));

% Find optimum frequency
[~,ib] = min(abs(S11));
[~,it] = max(abs(S21));

% Display values in Command Window
fprintf('\n MICROWAVE TWO-PORT SIMULATION\n');
fprintf('-------------------------------------\n');
fprintf('S11 at 10 GHz            = %.3f dB\n',S11dB(i0));
fprintf('S21 at 10 GHz            = %.3f dB\n',S21dB(i0));
fprintf('Return Loss              = %.3f dB\n',RL(i0));
fprintf('Insertion Loss           = %.3f dB\n',IL(i0));
fprintf('Transmission Efficiency  = %.2f %%\n',Eff(i0));
fprintf('Best Operating Frequency = %.2f GHz\n',f(ib));
fprintf('Maximum S21 Frequency    = %.2f GHz\n',f(it));

fprintf('\n VALUES AT BEST FREQUENCY\n');
fprintf('S11                      = %.3f dB\n',S11dB(ib));
fprintf('S21                      = %.3f dB\n',S21dB(ib));
fprintf('Return Loss              = %.3f dB\n',RL(ib));
```

```
fprintf('Insertion Loss            = %.3f dB\n',IL(ib));
fprintf('Transmission Efficiency   = %.2f %%\n',Eff(ib));

% Plot all six parameters
figure('Color','w');

subplot(2,3,1);
plot(f,S11dB,'b','LineWidth',1.5);
title('S11'); xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)'); grid on;

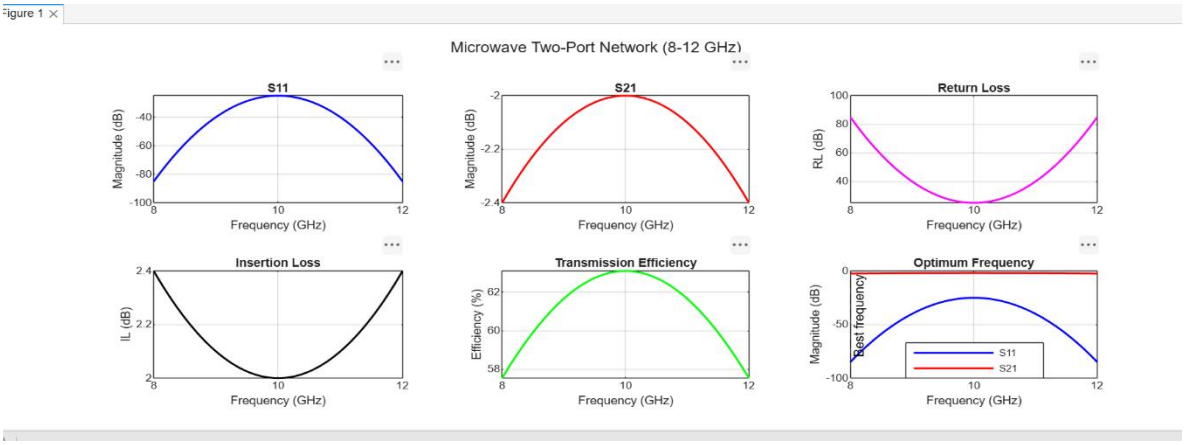
subplot(2,3,2);
plot(f,S21dB,'r','LineWidth',1.5);
title('S21'); xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)'); grid on;

subplot(2,3,3);
plot(f,RL,'m','LineWidth',1.5);
title('Return Loss'); xlabel('Frequency (GHz)');
ylabel('RL (dB)'); grid on;

subplot(2,3,4);
plot(f,IL,'k','LineWidth',1.5);
title('Insertion Loss'); xlabel('Frequency (GHz)');
ylabel('IL (dB)'); grid on;
subplot(2,3,5);
plot(f,Eff,'g','LineWidth',1.5);
title('Transmission Efficiency');
xlabel('Frequency (GHz)');
ylabel('Efficiency (%)'); grid on;
subplot(2,3,6);
plot(f,S11dB,'b','LineWidth',1.5); hold on;
```

```
plot(f,S21dB,'r','LineWidth',1.5);
xline(f(ib),'k--','Best frequency');
title('Optimum Frequency');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
legend('S11','S21','Location','best');
grid on;
sgtitle('Microwave Two-Port Network (8-12 GHz)');
```

5.Output:

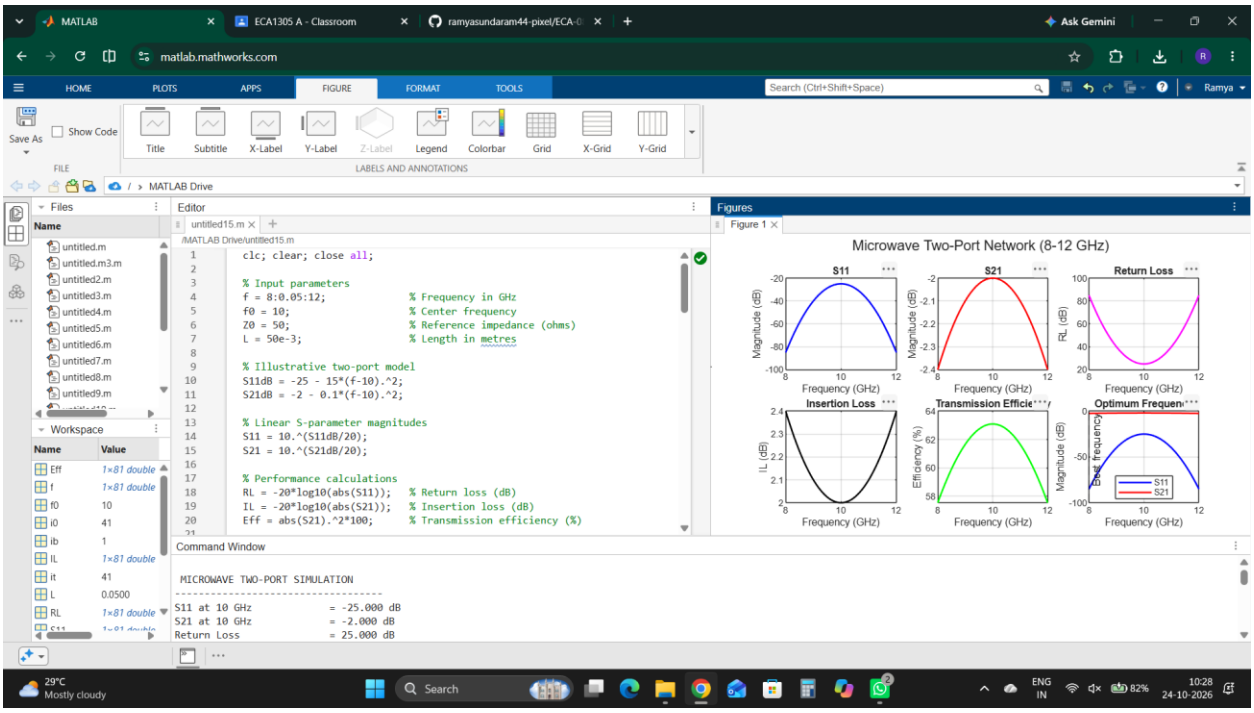



```
 MICROWAVE TWO-PORT SIMULATION
----------------------------------
S11 at 10 GHz              = -25.000 dB
S21 at 10 GHz              = -2.000 dB
Return Loss                = 25.000 dB
Insertion Loss             = 2.000 dB
Transmission Efficiency    = 63.10 %
Best Operating Frequency   = 8.00 GHz
Maximum S21 Frequency      = 10.00 GHz

 VALUES AT BEST FREQUENCY
S11                        = -85.000 dB
S21                        = -2.400 dB
Return Loss                = 85.000 dB
Insertion Loss             = 2.400 dB
Transmission Efficiency    = 57.54 %
>>
```

6.Result:

The microwave transmission line was modeled using MATLAB, and its scattering parameters were evaluated over the 8–12 GHz frequency range. Return loss, insertion loss, transmission efficiency, and propagation constant were calculated and plotted to study the transmission characteristics of the two-port network**.**



Simulation - 2

$f_0 = 10\ GHz$

$Z_0 = 50\ \Omega$

frequency sweep : 8 - 12 GHz

$L = 50mm$

$S_{11} = -25 - 15(f-10)^2$

$= -25 - 15[10-10]^2$

$S_{11} = -25\ dB$

$S_{21} = -2 - 0.1(f-10)^2$

$= -2 - 0.1(10-10)^2$

$S_{21} = -2\ dB$

Return loss :

$RL = -20\log|S_{11}|$

$|S_{11}| = 10^{-25/10} = 0.05623$

$RL = -20\log(0.0562)$

$= 25\ dB$

Insertion loss :

$IL = -20\log_{10}|S_{21}|$

$|S_{21}| = 10^{-2/20}$

$|S_{21}| = 0.7943$

$IL = -20\log(0.7943)$

$IL = 2\ dB$

Transmission efficiency :

$\eta = |S_{21}|^2 \times 100\%$

$= (0.7943)^2 \times 100\%$

$\eta = 63.10\%$

Best operating frequency :

$S_{11} = -25 - 15(f-10)^2$

At 8 GHz :

$S_{11} = -25 - 15(8-10)^2$

$= -25 - 15(4)$